

Asher Cronin (axe 1541)

CSD 5233 - HW 4

[illegible]

1.6 same insertions as 1.a, table size: 20, double hashing: $7 - (x \bmod 7)$

[illegible]

2.a insert 12, 18, 25, 42, 38, 17 into a size 10 table

-	-	12	42	-	25	-	17	18	38
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2.b rehash to size 20

-	42	-	25	-	-	-	-	12	-	-	-	17	18	38
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3. Linear probing: Simple to implement, increases index by 1 if collision, until there's an empty space. This results in a lot of clustering because similarly hashed keys will all end up in the same area. Because of the clustering, under high load factors, the time complexity gets closer to $O(n)$ than $O(1)$.

Quadratic Probing: Slightly more complex, the index first increases by 1 and 2 but then grows quadratically. ~~=====~~ This, to some extent, fixes the clustering problem, but not completely. Because it is less prone to clustering, its performance under high load factors is better than linear probing.

Separate Chaining: takes a different approach to dealing with collisions. Instead of finding an open space it just appends keys with the same index to an array or linked list. Searching is less optimized, but insertion will always be $O(1)$ with separate chaining.